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| Started on | 2023-09-09 19:02 |
| State | Finished |
| Completed on | 2023-09-09 19:12 |
| Time taken | 9 mins 24 secs |
| Grade | **9.00** out of 10.00 (**90**%) |

Question **1**
Correct
Mark 1.00 out of 1.00

Among the equations below choose an ordinary differential equation of 3rd order.

Select one:

- $\frac{d^2h}{dt^2} = t^3$
- $3\frac{d^2y}{dx^2} + \frac{dy}{dx} = \sin x$
- $\frac{\partial^2 u}{\partial t^2} - c^2\frac{\partial^2 u}{\partial x^2} = 0$
- $2y''' - 4y'' + y' - y = 0$ ✓

Your answer is correct.

The correct answer is: $2y''' - 4y'' + y' - y = 0$

Question **2**
Correct
Mark 1.00 out of 1.00

Among the equations below choose a partial differential equation of 2rd order.

Select one:

- $\frac{d^2h}{dt^2} = t^3$
- $3\frac{d^2y}{dx^2} + \frac{dy}{dx} = \sin x$
- $\frac{\partial^2 u}{\partial t^2} - c^2\frac{\partial^2 u}{\partial x^2} = 0$ ✓
- $2y''' - 4y'' + y' - y = 0$

Your answer is correct.

The correct answer is: $\frac{\partial^2 u}{\partial t^2} - c^2\frac{\partial^2 u}{\partial x^2} = 0$

Question 3
Incorrect
Mark 0.00 out of 1.00

Among the equations below choose a linear ordinary differential equation.

Select one:

- $y\frac{dy}{dx} + x = 0$ ✗
- $\frac{dy}{dt} - \cos y \sin t = \cos x$
- $\frac{d^2y}{dx^2} + y^2 = x$
- $t^3\frac{dx}{dt} - x = \sin t$

Your answer is incorrect.

The correct answer is: $t^3\frac{dx}{dt} - x = \sin t$

Question 4
Correct
Mark 1.00 out of 1.00

Choose the differential equation, which describes the radioactive decay, if it is known that the rate of decay is proportional (with constant $k > 0$) to the amount of radioactive substance ($A > 0$) present.

Select one:

- $\frac{dA}{dt} = -kA$ ✓
- $\frac{dA}{dt} = kA$
- $A = -k\frac{dA}{dt}$
- $A = k\frac{dA}{dt}$

Your answer is correct.

The correct answer is: $\frac{dA}{dt} = -kA$

Question 5
Correct
Mark 1.00 out of 1.00

Which variable is independent variable in the ordinary differential equation $\frac{d^2x}{dt^2} + a\frac{dx}{dt} + kx = 0$?

Select one:

- $t$ ✓
- $x$
- $a$
- $k$

Your answer is correct.

The correct answer is: $t$

Question 6

Correct

Mark 1.00 out of 1.00

Which variable is dependent variable in the differential equation $\frac{\partial u}{\partial x} - \frac{\partial u}{\partial y} = x - 2y$ ?

Select one:

- $u$ ✓
- $x$
- $y$
- $x - 2y$

Your answer is correct.

The correct answer is: $u$

Question 7

Correct

Mark 1.00 out of 1.00

Choose a differential equation that fits the physical description: The velocity at time $t$ of a particle moving along a straight line is proportional to the fourth power of its position $x$ with the coefficient of proportionality $k$.

Select one:

- $\frac{dx}{dt} = kx^4$ ✓
- $\frac{d^2x}{dt^2} = kx^4$
- $x^4 = k\frac{dx}{dt}$
- $\frac{d^4x}{dt^4} = kx^4$

Your answer is correct.

The correct answer is: $\frac{dx}{dt} = kx^4$

Question 8
Correct
Mark 1.00 out of 1.00

Among the expressions below choose a solution to the equation $y'' - y' - 2y = 2 - 2x - 2x^2$.

Select one:

- $y = 5e^{-x} - 2e^{2x} + x^2$ ✓
- $y = x^2 - x + 2e^{-x}$
- $y = 5e^{2x} - x^2$
- $y = \ln x - 18x^2 + x$

Your answer is correct.

The correct answer is: $y = 5e^{-x} - 2e^{2x} + x^2$

Question 9
Correct
Mark 1.00 out of 1.00

Which of the functions below is a solution to the initial value problem $\frac{d^2y}{dx^2} + y = 0, \quad y(0) = -1, \quad \frac{dy}{dx}(0) = 1?$

Select one:

- $y = \sin x - \cos x$ ✓
- $y = x^4 - 2x^2 + x - 1$
- $y = e^x - 1$
- $y = (x - 1)\cos x$

Your answer is correct.

The correct answer is: $y = \sin x - \cos x$

Question 10
Correct
Mark 1.00 out of 1.00

From the equations below choose a separable differential equation.

Select one:

- $\frac{dy}{dx} = \frac{2x+xy}{y^2+1}$ ✓
- $\frac{dy}{dx} = 1 + xy$
- $\frac{dy}{dx} = \sin(x + y)$
- $\frac{dy}{dx} = y^2 + x^2$

Your answer is correct.

The correct answer is: $\frac{dy}{dx} = \frac{2x+xy}{y^2+1}$